

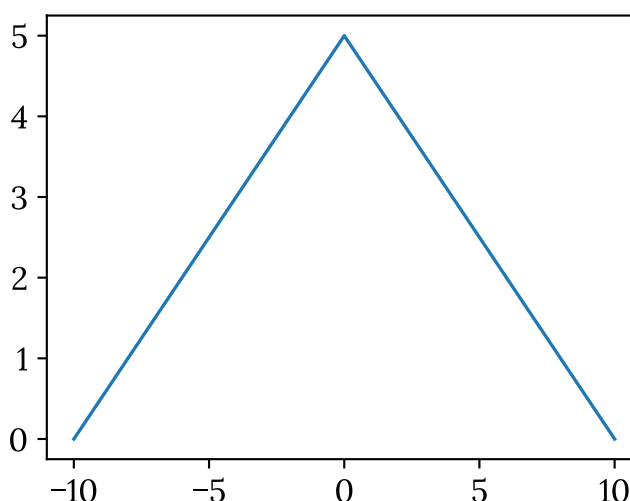
CNS2025: Homework 16

Due: 2025-12-24 23:59

Perform stochastic gradient descent with learning rate $\epsilon_w = 1$ on a set of basis neurons with tuning curves

$$f_b(s) = \exp\left(-\frac{(s - s_b)^2}{2}\right),$$

where the preferred stimulus values are $s_b = -10, -8, \dots, 8, 10$. Use the triangle function:



as the target function. Draw the approximated function after 0, 20, and 100 steps.

Show the components of output in each basis function.

Note: This is similar to what is done with cosine target function in the [code16.ipynb](#) code file. You can copy relevant code from the file and adapt it to the triangle target function here.